

Namrata Roy

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Important links

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NASA ADS Library: [Namrata Roy's Publications](#)

Website: <https://namrataroy.github.io/>

Google Scholar: [Namrata Roy's Profile](#)

Summary and Highlights

- **Research Focus:** Extragalactic astrophysics, galaxy evolution with multi-wavelength observations, black holes and active galactic nuclei feedback, baryon cycle in interstellar and circumgalactic medium, star formation quenching, relativistic jets, Epoch of Reionization and the escape of Lyman-continuum ("first ionizing") photons in the early Universe, and theoretical modeling.
- **Publications:** 62 refereed papers (11 first-author), 1600+ citations, h-index=23
- **Funding and telescope proposals:** Won ~ 1.1M USD in funding grants, PI (or deputy-PI) on 10 different highly competitive telescope programs, co-I in 15+ additional programs, obtained >220 hrs HST+JWST telescope, >30 hrs ground-based telescope time
- **Talks/Presentations:** 47 talks, 18 invited.
- **Students mentored:** 3 High School students, 3 Bachelors, 1 Masters, and 2 PhD students.

Academic Positions

- **Raman Research Institute** **Bangalore, India**
Assistant Professor *August 2026 -*
- **Arizona State University** **Tempe, AZ**
Exploration Prize Postdoctoral Fellow (Independent fellowship) *August 2025 - June 2026*
Independent Fellowship, collaborators: Prof. Sanchayeeta Borthakur, Prof. Rogier Windhorst, Prof. Allison Noble
- **Johns Hopkins University** **Baltimore, MD**
Post-Doctoral Fellow *September 2022 - July 2025*
Collaborators: Prof. Timothy Heckman & Dr. Alaina Henry
- **University of California, Santa Cruz** **Santa Cruz, CA**
Osterbrock Scholar *September 2019- June 2022*
Advisor: Prof. Sandra Faber & Prof. Robert Williams
- **Harish Chandra Research Institute** **Allahabad, India**
Summer Research Fellow *2015*
Advisor: Prof. Tapash Kumar Das

Other Visiting Positions

- **Center for Computational Astrophysics, Flatiron Institute** **New York City, NY**
Visiting Scientist *September 2021*

Collaborators: Prof. Rachel Somerville & Prof. Chris Hayward

- **Space Telescope Science Institute (STScI)**
Visiting Scientist
Collaborators: Dr. Alaina Henry, Dr. Kate Rowlands, &
Dr. Katherine Alatalo

Baltimore, MD
September 2022-July 2025

Education

- **University of California, Santa Cruz**
PhD, Astronomy/Astrophysics (Supervisor: Prof. Kevin Bundy)
Thesis: Star formation suppression and feedback in nearby galaxies
Santa Cruz, CA
August 2016 - June 2022
- **Presidency University**
M.Sc., Physics (Thesis Supervisor: Prof. Durgesh Tripathi, IUCAA)
Thesis: Characteristics of Chromospheric Spectral Line in Solar Flares
Kolkata, India
2014-2016
- **Presidency University**
B.Sc., Physics (Project Supervisor: Prof. Ritaban Chatterjee)
Paper: Probing the jets of blazars using the temporal symmetry of their multiwavelength outbursts
Kolkata, India
2011-2014

Research Interests

- **AGN Feedback & Galaxy Evolution (Extragalactic Observations)**
 - Role of active galactic nuclei (AGN) in regulating star formation and driving galaxy evolution
 - Physical mechanisms behind star formation quenching
- **Gas Flows in Galaxies**
 - Multi-phase gas in galaxies (ionized, neutral, and molecular)
 - Outflows and inflows in the interstellar medium (ISM)
 - Circumgalactic medium (CGM) gas characterization and its connection to galactic gas cycle
- **Radio Galaxies & High-Energy Astrophysics**
 - Jet-ISM-CGM interactions in large-scale collimated radio-jetted galaxies
 - Feeding and feedback in low-power, compact (non-jetted) radio AGNs
 - Time variability of Blazar flares
- **Cosmic Reionization and High-redshift Galaxies**
 - Ly α emitters as probes of galaxies near the Epoch of Reionization
 - Lyman continuum escape fraction and its role in cosmic reionization
- **Data Science and Novel Observational Strategies**
 - Multi-wavelength UV–Optical-IR-radio imaging, high resolution and spatially resolved spectroscopy (IFU, MOS, Echelette) from ground and space based observatories
 - Harnessing big data from large surveys to derive population trends and statistics to compare with theoretical interpretation
 - Development of next-generation instrumentation
- **Theoretical Modeling and Computational Astrophysics**
 - Biconical outflow and disk rotation modeling
 - High resolution relativistic hydrodynamic jet simulations
 - Cosmological zoom-in simulations (FIRE)

Successful Proposal Awards on International Facilities

PI/deputy-PI/co-I on 25+ programs across HST, JWST, Magellan, Keck, MMT, GMRT, VLT, LBT and ALMA.

PI and deputy-PI Proposals.....

- **2026: LBT MODS 2026: Director's discretionary time (PI)**
Probing the quasar-radio galaxy pair
Awarded 0.5 night of LBT time
- **2026: MMT Binospec IFU 2026: Director's discretionary time (PI)**
Where Radio jets meet the halo gas
Awarded 0.5 night of Binospec time
- **2025: Magellan Baade FIRE 2026A: ID 357 (PI)**
Quiet Galaxy, Large-scale Winds: Warm molecular gas and shocks in massive, quiescent "red geyser" galaxies
Awarded 1 night of Magellan time
- **2025: HST Cycle: 33 GO_18096 (PI)**
COS-RAD: Where Radio Jets Meet the Halo gas
Awarded 26 orbits of HST time
- **2025: GMRT 49_066 (PI)**
Revealing Aged Radio Plasma and AGN Feedback in "Red Geysers": old-and-dead galaxies with low luminosity AGNs and large scale biconical outflows
Awarded 4 hours of GMRT time
- **2025: MMT/Binospec ID: 331 (PI)**
Jet-ISM coupling along the radio lobes in four nearby Radio galaxies
Awarded 2 nights of MMT time
- **2025: LBT 2025A (PI)**
Exploring the Physics of a New Class of Lyman-Leaky Galaxies
Awarded 1 night of LBT time
- **2024: Magellan Clay LLAMAS 2024A: ID 54 (deputy-PI)**
Mapping Outflows from Starbursts into the Circum-Galactic Medium
Awarded 1 night of Magellan time
- **2023: HST Cycle: 31 GO_17485 (PI)**
Dissecting Red geyser winds: low luminosity AGNs with large scale outflows in the ionized phase
Awarded 5 orbits of HST time
- **2022: Keck OSIRIS 2022B_U051 (science PI/ deputy-PI)**
Probing feedback in the nuclear region of low luminosity AGN host "red geysers"
Awarded 2 nights of Keck time
- **2020: APEX P105-105.20EE (deputy-PI)**
Cold gas on the red sequence: maintenance mode feedback in action
- **2019: ALMA 2019.1.00597.S (deputy-PI)**
Cold gas on the red sequence: resolving maintenance-mode feedback in action

Co-I Proposals.....

- **2026: JWST Cycle: 5 GO_10208**
Resolving the growth and evolution of galaxies across cosmic noon with JWST slit-stepping spectroscopy
Awarded 115 hours of JWST time
- **2026: JWST Cycle: 5 GO_10815**
Following the Shocks: JWST IFU Spectroscopy of NGC 4258's Radio Arcs
Awarded 15 hours of JWST time
- **2025: VLT/ERIS: P116**
Ionized gas in quiescent galaxies: the key to quenching?
- **2025: JWST Cycle: 4 GO_7041**
The CLASSYIR Treasury: Unveiling the Cosmic Engines Powering Galaxies with JWST/MIRI
Awarded 78 hours of JWST time
- **2025: JWST Cycle: 4 GO_7076**
A comprehensive population study of Little Red Dots: Connecting early BH and galaxy growth
Awarded 87 hours of JWST time
- **2024: JWST Cycle: 3 GO_5293**
Galactic Winds in the Early Universe: observing outflows in emission and absorption in a typical $z \sim 6$ galaxy
Awarded 10 hours of JWST time
- **2024: JWST Cycle: 3 GO_6074**
The First Measurement of AGN Feedback in Action in the First Billion Years
Awarded 8 hours of JWST time
- **2024: XRISM Cycle: 1 Regular**
Quantifying AGN Feedback in the Archetypical powerful Radio Galaxy Cygnus A
- **2024: HST Cycle: 32 AR_17859**
Mocking Galactic Winds: A new approach to constraining feedback by modeling the UV spectra of galaxies
Archival HST proposal
- **2024: Keck OSIRIS 2024B**
Using $\text{Pa-}\alpha$ as an outflow tracer in CLASSY star-forming galaxies with outflows
- **2023: HST Cycle: 31 AR_17559**
Galactic Winds Unveiled: Leveraging Cloud Simulations with Radiative Transfer to Constrain Feedback
Archival HST proposal
- **2023: HST Cycle: 30 AR_17042**
Are Galactic Outflows Seen in Absorption and Emission Lines Tracing the Same Gas?
Archival HST proposal
- **2020: GBT 19B-336**

Exploratory Observations of CO(1-0) 115.271 GHz Emission in MaNGA Galaxies

- **2019: GMRT 36_022**
A Study of Radio Mode Feedback in Red Geysers from SDSS IV's MaNGA Survey
- **2018: Keck KCWI 2018B_U081**
Towards the first measurement of gas-phase metallicity in early type LINER galaxies
- **2018: UCO Grant**
Modeling Fiber Performance for Ultra-faint Spectroscopy
- **2018: NSF Proposal A18_0759**
Red geysers and the suppression of star formation

Awards and Honors

- **2025: Exploration Prize Postdoctoral Fellowship**, School of Earth and Space Exploration, Arizona State University
One highly competitive postdoctoral fellowship awarded annually across Astrophysics, Cosmology, Planetary Science, Geology, and Earth Sciences.
- **2022: Space Telescope Science Institute (STScI) Prize Fellowship** (*offer declined*)
Four-year independent fellowship at STScI awarded on an international, highly competitive basis.
- **2021: UC Dissertation Year Chancellor's Fellowship**
University-wide, merit-based fellowship awarded on a competitive basis to outstanding doctoral students in the final stages of their dissertations.
- **2019–2022: AAS National Osterbrock Leadership Program Fellowship**, UC Santa Cruz
Awarded to graduate students demonstrating exceptional leadership potential and commitment to community-building in astronomy.
- **2015: Summer Research Fellowship**, Harish-Chandra Research Institute, India
Selected for the prestigious Visiting Students Program in theoretical and observational astrophysics.
- **2011–2016: INSPIRE Scholarship**, Department of Science & Technology, Government of India
National scholarship awarded to the top $\sim 1\%$ of students pursuing careers in the basic sciences.
- **2022: Waterloo Centre for Astrophysics Prize Postdoctoral Fellowship** (*offer declined*)
Three-year prize fellowship offered to a small cohort (typically two fellows per year).

Funding Awards and Grants

Total external support: approximately \$1.1 million USD, including $\sim$ \$430k from JWST, $\sim$ \$220k from HST, $\sim$ \$160k from NSF, plus competitive fellowships.

- Hubble Space Telescope ($\sim$ \$220,500 USD)
 - PI programs: 17485, 18096

- Co-I programs: 17859, 17559, 17042
- James Webb Space Telescope (~ \$430,000 USD)
 - Co-I programs (large): 7041, 7076 (>170+ hours)
 - Co-I programs (small): 5293, 6074 (~ 18 hours)
- National Science Foundation (~ \$160,000 USD)
 - Co-I program: 0759
- National Osterbrock Leadership Program Fellowship (~ \$15,000 USD)
- UC Chancellor's Dissertation Year Fellowship (~ \$52,000 USD)
- Exploration Prize Postdoctoral Fellowship (~ \$255,000 USD)
- Conference organization funds, STScl Spring Symposium
"Recipes to Regulate Star Formation at All Scales: From the Nearby Universe to the First Galaxies"
 (~ \$10,000 USD)

Seminars & Conference presentations:

47 talks (18 invited); Watch selected presentations [here](#).

Invited Talks.....

- **October 2026:** (*Scheduled*) WAVES (Widening Astrophysics with Voices, Equity, and Science) meeting, International Centre for Theoretical Sciences (ICTS – TIFR), **Invited Talk**.
- **September 2026:** (*Scheduled*) Cosmic Connections: AI/ML Across Cosmic Scales, The Institute of Mathematical Sciences, **Plenary Talk**.
- **April 2026:** NASA Galaxies Science Interest Group (SIG) seminar, **Invited Seminar**.
- **March 2026:** Cosmology Seminar, Arizona State University, **Invited Seminar**.
- **February 2026:** AGN & Energy Flows (**Invite only**) Workshop, Flatiron Center for Computational Astrophysics
- **January 2026:** NASA AGN Science Interest Group (SIG) seminar series, **Invited Seminar**.
- **June 2025:** JWST at the Gates of Cosmic Dawn (Meeting in honor of Garth Illingworth), Johns Hopkins University, **Invited Talk**.
- **March 2024:** CTC seminar, University of Maryland College Park, **Invited Colloquium**.
- **August 2023:** 2023 Santa Cruz Galaxy Conference, **Invited Talk**.
- **August 2022:** Colloquium at the School of Astrophysics, Presidency University, India, **Invited Colloquium**
- **August 2022:** Colloquium at the Raman Research Institute, India, **Invited Colloquium**
- **August 2022:** Colloquium at the Indian Institute of Astrophysics, India, **Invited Colloquium**.
- **August 2022:** Colloquium at the Indian Institute of Science Education and Research Kolkata (IISER-Kolkata), India, **Invited Colloquium**.
- **July 2022:** Colloquium at the Indian Institute of Science Education and Research Pune (IISER-Pune), India, **Invited Colloquium**.
- **June 2022:** SDSS-MaNGA Scicon3, Virtual, **Invited talk**.
- **September 2021:** Colloquium at Harvard Center for Astrophysics (CfA), **Invited Colloquium**.
- **August 2020:** Alumni Lecture Series, Presidency University, India (Virtual), **Invited lecture**
- **Nov 2019:** Colloquium at University of California Berkeley, USA, **Invited talk**

Contributed Talks.....

- **March 2026:** End of Star Formation Conference, University of Illinois
- **January 2026:** 247th AAS meeting
- **August 2025:** IAU General Assembly Talk

- **June 2025:** AAS Conference Talk
- **September 2024:** UT Austin ExGal Seminar
- **September 2024:** Steward / NOIRLab FLASH Talk
- **August 2024:** IAU Symposium
- **October 2023:** 2023 JSI Conference: Winds throughout the Universe
- **July 2023:** 2023 National Astronomy Meeting (gave 2 oral presentations on 2 different topics)
- **June 2023:** First Light Conference, MIT
- **July 2022:** Colloquium at the Inter-University Centre for Astronomy and Astrophysics, India
- **July 2022:** Colloquium at the National Centre for Radio Astrophysics - Tata Institute of Fundamental Research, India
- **June 2022:** UCSC Graduate Research Symposium
- **May 2022:** University of California Santa Cruz Public Talk
- **November 2021:** Steward / NOIRLab Galaxy Group meeting
- **September 2021:** UC Berkeley Thursday short talk series
- **September 2021:** Yale Galaxy lunch
- **August 2021:** SDSS Collaboration meeting
- **July 2021:** SDSS-MaNGA Scicon1
- **November 2020:** Center for Computational Astrophysics, Flatiron, Friday Seminar
- **October 2020:** Young Astronomers on Galactic Nuclei (yAGN) meeting, 2020
- **October 2020:** MaNGA Collaboration meeting
- **September 2020:** Keck Science meeting, 2020
- **May 2020:** SDSS Collaboration meeting
- **March 2019:** MaNGA Collaboration meeting
- **August 2018:** Santa Cruz Galaxy Conference
- **July 2018:** Supermassive black holes and their host galaxies, Asiago Astrophysical Observatory, University of Padua
- **March 2018:** MaNGA Collaboration meeting
- **June 2016:** Astronomical Society of India (ASI) meeting

International Scientific Collaborations

Leading roles in several large, international collaborations, ensuring a steady stream of frontier datasets and collaborative opportunities for students in my group.

- **JWST Collaborations**
 - MSA-3D (59 hours of JWST): AGN/outflow science group lead
 - COSMOS-Webb (> 250 hours of JWST): Research Group Member
 - GLASS-ERS (36 hours of JWST): Data Analyses lead, Multiple collaborations on Ly α emitters/high- z galaxy characterization
 - Cosmic Springs/MACS0308 (10 hours of JWST): Science Lead on gas flows, resolved ionization and metallicity characterization
 - CLASSY-IR (78 hours of JWST): ISM characterization
 - Obscured AGNs at high- z and Little Red Dots (90+ hours of JWST)
- **HST Collaborations**
 - COS-RAD (26 HST orbits): Science PI and data analyses lead

- Outflow modeling group (HST archival data): Role – Observational comparison
- **SDSS IV - MaNGA Collaboration:** Science lead on driving low-luminosity AGN science; led follow-up studies with multi-wavelength ancillary observations.
- **Subaru/PFS Collaboration:** Radio AGN/ outflows working group
- **LOFAR/LOTSS Collaboration:** AGN radio morphology science using LOFAR/MANGA/DESI
- **SDSS-V Collaboration:** AGN variability science
- **SPOGs Collaboration:** Shocked post-starburst galaxy group at STScI, interest in post-starburst galaxies, outflows.
- **GREENS Collaboration:** Galaxies at Reionization and Extreme Emission Sources (GREENS) group at STScI, science collaboration on reionization and extreme emission line sources

Teaching & Mentoring

I have mentored 9 students (3 high school, 3 Bachelor's, 1 Master's, 2 PhD), leading to student-authored papers, scholarships, and conference awards, and I have taught large introductory astronomy courses, Python training programs, and designed coursework for astronomy classes in India.

Mentoring.....

- **2024–Present:** Research mentor, Johns Hopkins University PhD student **Timothy Honablew**
Thesis topic: "Outflows in nearby Seyfert galaxies"
Student success: First paper in preparation (out of three planned publications). Timothy has completed his doctoral candidacy examination in February 2026.
- **2024–Present:** Primary research mentor, UC Santa Cruz undergraduate student **Arian Moghni**, Rowland Summer Research Fellowship
Topic: "AGN feedback in low-luminosity AGN host galaxies"
Student success: First student led paper accepted. Second paper in preparation. Arian won the prestigious Goldwater Scholarship, Ron Ruby Scholarship and the best poster award at the UC Santa Cruz APS poster presentation for this research. His work has been highlighted by multiple press releases.
- **2023–2025:** Primary mentor, Master's thesis student **Anuroop Dasgupta**, Presidency University, India (co-mentor: faculty of Presidency University – Dr. Ritaban Chatterjee)
Thesis topic: "Metallicity gradients in dwarf galaxies"
Student success: Master's thesis submitted. Student led paper submitted. Anuroop won the best speaker award at the West Bengal Regional Science Congress for presenting his thesis work. He is currently pursuing a PhD at Universidad Diego Portales, Chile.
- **2023–2024:** Mentor, Johns Hopkins University PhD student **Annie Gao**, JHU Physics & Astronomy mentoring program
Mentoring focus: Guidance on graduate school, research, and career paths, with additional academic mentoring in ISM physics in the local galactic neighborhood.
- **2022:** Mentor, undergraduate student **Anand Sharma**, IISER Pune, India

Topic: "AGNs in passive elliptical galaxies"

- **2017:** Primary research mentor, Science Internship Program (three high school students: **Atirath Dhara, Kaela McConnell, Jurij Waite**), UC Santa Cruz
Topic: A study of galaxies and quasars in the background of the Andromeda Galaxy
Student success: Research presented and published at the 2018 AAS meeting.
- **2017:** Co-advisor, undergraduate research project of **Marina Huang**, UC Santa Cruz (primary mentor: Prof. Kevin Bundy)
Topic: Development of an automatic algorithm to identify MaNGA red geysers
Student success: Marina is currently employed as an engineer at an optical and photonics products manufacturer in the San Francisco Bay Area.

Teaching.....

- **2021:** Guest instructor for the LAMAT Summer Research Training Program, UC Santa Cruz
Conducted Python workshops, hands-on tutorials in astronomical data analysis, and demonstrations of research tools for a cohort of ~10 students.
- **2020–2023:** Designed the introductory astronomy curriculum, Python training modules, course structure, reading materials, and short thesis project ideas for the newly established Astrophysics Master's program at the School of Astrophysics, Presidency University, in close collaboration with faculty there.
- **2018:** Teaching Assistant, UC Santa Cruz (class of ~150 undergraduates)
ASTR 5: Overview of the Universe
Led discussion sections, designed homework assignments and worksheets, and held problem-solving tutorial sessions.
- **2017:** Teaching Assistant, UC Santa Cruz (class of ~150 undergraduates)
ASTR 2: The Formation and Evolution of the Universe
Held office hours, graded examinations, and led problem-solving tutorial sessions.

Press Coverage

- [How quiet galaxies stay quiet: cool gas feeds black holes in 'red geysers'](#), UC Santa Cruz News, January 8, 2026
- ["Red Geyser" Galaxies Have Plenty of Star-Forming Gas But Don't Form Stars](#), Universe Today, January 30, 2026
- [Webb Watches the Most Distant Galactic Merger Ever Seen](#), Universe Today, March 24, 2024
- [Detailed pictures show galaxy growth in the early universe was much faster than first thought](#), Physics.org, March 12, 2024
- [Astrophysicists confirm the faintest galaxy ever seen in the early universe](#), UCLA Newsroom, May 31, 2023
- [Small-Scale Radio Jets in Red Geyser Galaxies](#), NRAO News, January 20, 2022

Professional Service and Leadership

Telescope and Grant Review.....

- **2025:** Invited review panelist, **National Science Foundation** Astronomy and Astrophysics Grants (NSF–AAG)
Goal: Evaluate and rank research proposals to allocate funding to US university faculty
- **2025:** Invited Time Allocation Committee (TAC) panelist, **James Webb Space Telescope** (JWST)
Goal: Allocate highly competitive telescope time based on proposal merit (oversubscription > 10:1)
- **2024:** Invited Time Allocation Committee (TAC) panelist, **Hubble Space Telescope** (HST)
Goal: Allocate highly competitive telescope time based on proposal merit (oversubscription ~ 8:1)
- **2025:** Invited proposal reviewer, **Giant Metrewave Radio Telescope** (GMRT)
Goal: Review and rank requests for GMRT observing time based on scientific merit
- **2020–Present:** Referee for **manuscripts** in *MNRAS* and the *AAS* journals.

Community Planning and Strategic Reports.....

- **2024–2025:** Core group member, **Habitable Worlds Observatory (HWO)** AGN Science Planning Committee
Goal: Define AGN science goals and requirements for future mission (arXiv:2508.01437)
- **2024:** Invited contributor, **Astronomical Society of India (ASI) Vision Document**
Goal: Recommend national priorities for galaxies, clusters, and the intergalactic medium for the coming decade (2025JApA...46...32K)
- **2023–2024:** Member, **NASA Science Analysis Group** “Astrophysics with Equity: Surmounting Obstacles to Membership”
Goal: Analyze diversity, equity, and inclusion in NASA astrophysics and contribute to a community white paper
- **2018–2019:** Member, **TMT/WFOS** (early proposed instrument) Planning and Initial Development Team, UC Santa Cruz
Goal: Contribute to the initial optical multi-object spectrograph design for the Thirty Meter Telescope

Conference, Colloquium, and Program Organization.....

- **2025–Present:** Organizer, Beus Seminar Series, Arizona State University
- **2024–2025:** Organizer, weekly AGN & Galaxies Journal Club, Space Telescope Science Institute (STScI)
- **2024–2025:** Organizer, joint Johns Hopkins University (JHU) & STScI weekly colloquium
- **April 2024:** Organizer and Scientific Organizing Committee member, STScI Spring Symposium “Recipes to Regulate Star Formation at All Scales: From the Nearby Universe to the First Galaxies”

(200+ participants)

- **2023–2024:** Co-creator, CARINAs Colloquium Series
Goal: Highlight astronomy research led by Indian women astronomers
- **2023–2024:** Organizer, JHU bi-weekly AstroCoffee discussions
- **2020:** Organizer, “Career Beyond Academia” panel (Osterbrock Program, UC Santa Cruz)
Goal: Connect UCSC astrophysics PhD students with alumni in industry and data science careers
- **2019–2021:** Organizer, leadership events as an Osterbrock Fellow, UC Santa Cruz
Goal: Run a mini-grants program for student-led leadership projects and host discussion panels with faculty and alumni

Public Engagement.....

- **2021:** Invited panelist, *Presision* undergraduate symposium, Presidency University, India
Goal: Lead discussion on academic and career pathways to undergraduate physics students
- **2021:** Mentor, Society of Physics Students and Women in Physics & Astronomy
Goal: Lead peer-mentoring efforts by pairing each woman undergraduate student with a graduate student mentor to support research development and career planning
- **2016–2017:** Lead, “Ask-an-Astronomer” outreach program, UC Santa Cruz
Goal: Answer astronomy questions from the public and engage broader audiences in astrophysical research

References Details

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- **Prof. Timothy Heckman**, Dept. of Physics & Astronomy, Johns Hopkins University
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Summary of Publications

- Refereed Publications: **62**
- First-authored refereed publications: **11**, Second-authored refereed publications: **3**
- Student-led publications: **1** accepted, **1** under review, **1** in prep
- Total number of Citations: **1609**
- h-index: **23**

Refereed Publications:

First & Second Author Publications: († means student paper)

1. **2026:** Roy, N., Borthakur, S., Heckman, T., Singh, T. 2026, *Lighting Up the CGM: Strong, Jet-Aligned H α Emission around Radio Galaxies*, arXiv:2606.30088, Under review in ApJL
2. **2026:** Roy, N., Heckman, T., Henry, A. 2026, *Mapping Jet-Gas Coupling and Energetic Ionized Outflows in High-Redshift Radio Galaxies with JWST/NIRSpec*, ApJ, 1002, 57R
3. **2025:** Roy, N., Henry, A., et al. 2025, *MSA-3D: Uncovering Weak AGNs and Resolved Outflows in Disguise in $z \sim 1$ Star-Forming Galaxies*, arXiv:2510.11326, Under Review in ApJ
4. **2025:** Roy, N., Heckman, T., Henry, A. et al. 2025, *Lyman continuum leakage from massive leaky starbursts: A different class of emitters?*, ApJ, 992, 91
5. **2024:** Roy, N., Heckman, T., et al. 2024, *JWST Reveals Powerful Feedback from Radio Jets in a Massive Galaxy at $z = 4.1$* , ApJ, 970, 69
6. **2023:** Roy, N., Henry, A., Heckman, T., et al. 2023, *Early Results from GLASS-JWST XXII. Rest-frame UV-Optical Spectral Properties of Ly α Emitting Galaxies at $3 < z < 6$* , ApJL, 952, 14
7. **2021:** Roy, N., Moravec, E., Bundy, K., et al. 2021, *Radio Morphology of Red Geysers*, ApJ, 922, 230
8. **2021:** Roy, N., Bundy, K., et al. 2021, *Signatures of inflowing gas in red geyser galaxies hosting radio-AGN*, ApJ, 919, 145
9. **2021:** Roy, N., Bundy, K., et al. 2021, *Evidence of Wind Signatures in the Gas Velocity Profiles of Red Geysers*, ApJ, 913, 33
10. **2019:** Roy, N., Chatterjee, R., et al. 2019, *Probing the jets of blazars using the temporal symmetry of their multiwavelength outbursts*, MNRAS, 482, 743
11. **2018:** Roy, N., Bundy, K., et al. 2018, *Detecting Radio AGN Signatures in Red Geysers*, ApJ, 869, 117
12. †**2026:** Moghni, A., Roy, N., et al. 2026, *Galactic Rain: Cold Gas in Red Geyser Galaxies and Their Connection to Radio Activity and Interactions*, ApJ, 1003, 31 (Student Paper)
13. †**2026:** Dasgupta, A., Roy, N., Samui, S. et al. 2026, *Spatially Resolved Gas-Phase Metallicity Gradients in 382 MaNGA Low-Mass Galaxies: The Role of Concentration, Gas Velocity Dispersion, and Environment*, Under Review in A&A (Student Paper)

14. **2024:** Heckman, T., **Roy, N.**, et al. 2024, *Mergers, Radio Jets, and Quenching Star-Formation in Massive Galaxies: Quantifying their Synchronized Cosmic Evolution & Assessing the Energetics*, ApJ, 977, 125
15. **2025:** **Roy, N.**, Xu, X., Abdurro'uf, A. et al. 2025 (in prep), *Feedback in Action at Cosmic Dawn: Spatially Resolved Galactic Outflows at $z = 6.2$*

Co-authored Publications:

16. **2026:** Borthakur, S., et al. 2026 (incl **Roy, N**), *Revealing Cosmic Ecosystems with the Hubble Space Telescope in 2030s and Beyond*, White Paper submitted to HST STScI Call for White Papers 2026, arXiv:2606.11506
17. **2026:** Espejo, S. J., et al. 2026 (incl **Roy, N**), *MSA-3D: Rotation Curves and Dark Matter Fractions at $z \sim 0.5 - 1.7$ with JWST/NIRSpec*, submitted to A&A, arXiv:2606.27853
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